SC Algebra 1
1.1 – Numerical Expressions • **KEY**

Warm Up:

1. $(-1) + (-17)$

   → -18

2. $21 - 13 + (-4)$

   → $8 + (-4) = 4$

3. $-8 - (-7) - (-2)$

   → $-8 + 7 + 2 = 1$

4. $-5 - (-4) + 3$

   → $-5 + 4 + 3 = 2$

5. One winter day, the difference between the warmest and coldest temperatures in the US was 74 degrees. If the warmest temperature was 61 degrees, what was the coldest temperature on that day?

   → $61 - 74 = -13°$ (13° below zero)

Speak Like A Mathematician:

numerical expression: A combination of numbers and operations – no variables and no equal sign.

exponent: The small raised number telling how many times the base is used as a factor.

base: The number being multiplied repeatedly in a power.

evaluate: To find the value of an expression by carrying out all of its operations.

Example 1: Translate a Verbal Expression

- one plus eight divided by three
  - → $1 + 8 \div 3$
- seven times fifteen minus two times nine
  - → $7 \cdot 15 - 2 \cdot 9$ ($= 105 - 18 = 87$)
- twelve less than two times five
  - → $2 \cdot 5 - 12$ ($= -2$). “Less than” REVERSES the order – writing $12 - 2 \cdot 5$ is the classic error.

Example 2: Translate a Verbal Expression with Grouping

- the sum of five and nine divided by two
  - → $(5 + 9) \div 2 = 14 \div 2 = 7$
- eight less than the product of six and four
  - → $6 \cdot 4 - 8 = 16$. Note: this phrase needs no parentheses – the product happens first anyway. Its trap is the “less than” reversal.
- nine times the difference of 8 and 3
  - → $9(8 - 3) = 9(5) = 45$

Example 3: Write a Numerical Expression

- A person's handicap in bowling is usually found by subtracting the person's average from 200, multiplying by 2, and dividing by 3. Marley's bowling average is 170. Write a numerical expression for Marley's handicap in bowling.

  → $2(200 - 170) \div 3 = 2(30) \div 3 = 20$. The parentheses are REQUIRED – without them, $200 - 170 \cdot 2 \div 3$ is a different value.

- To convert a temperature in degrees Celsius to degrees Fahrenheit, multiply the temperature in degrees Celsius by 9, divide by 5, and add 32. Write a numerical expression to convert 37 degrees Celsius to Fahrenheit.

  → $37 \cdot 9 \div 5 + 32 = 333 \div 5 + 32 = 66.6 + 32 = 98.6°F$ (normal body temperature)

Example 4: Evaluate Expressions

- 2^4

  → $2 \cdot 2 \cdot 2 \cdot 2 = 16$. Base 2, exponent 4 – NOT $2 \cdot 4$.

- 4^5

  → $4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 = 1024$ ($16 \cdot 16 \cdot 4$ is a fast path)

Example 5: Order of Operations (GEMS)

GEMS: Grouping → Exponents → Multiply/Divide (left to right) → Subtract/Add (left to right). Grouping is more than parentheses – brackets, fraction bars, and absolute value bars all group.

- $20 - 7 + 8^2 - 7 \cdot 11$

  → exponent: 64; multiply: 77; then left to right: $20 - 7 = 13$, $13 + 64 = 77$, $77 - 77 = 0$

- $(4 + 5)^2 / 3(7 - 4)$

  → top: $(9)^2 = 81$; bottom: $3(3) = 9$; $81 \div 9 = 9$

- $15 - [10 + (3 - 2)^2] + 6$

  → innermost: $(1)^2 = 1$; $[10 + 1] = 11$; $15 - 11 + 6 = 10$

- $(2^3 - 5) / (15 + 9)$

  → top: $8 - 5 = 3$; bottom: 24; $3/24 = 1/8$

The fraction bar is a grouping symbol: it groups the ENTIRE numerator and the ENTIRE denominator. Writing $(2^3 - 5) \div 15 + 9$ is the most common error in this lesson.

Example 6: Write and Evaluate Expressions

- Thursdays are Student Days at LSC Theaters. Student tickets are \$5 and popcorn refills are free. You buy a ticket and a 20-ounce bottle of pop (\$5), and you will split the cost of a large tub of popcorn (\$7.50) and large candy (\$3.50) with a friend. Write an expression, then evaluate it to find how much money you should bring.

  → $5 + 5 + (7.50 + 3.50) \div 2 = 10 + 11 \div 2 = 10 + 5.50 = \$15.50$

- While she was on a 7-day vacation, Ms. Hernandez boarded her dog at a kennel. If her boarding budget is $250, can Ms. Hernandez afford a daily extra walk and one wash and nail trimming? Explain.

Service	Cost
Board	$24 per day
Wash	$45
Extra Walk	$4 per day
Nail Trimming	$12
Vitamins	$1 per day

→ 24(7) + 4(7) + 45 + 12 = 168 + 28 + 45 + 12 = $253. NO – she is $3 OVER budget.

Tonight's Homework: Paper Practice 1.1 (#1–12)